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# Multiple Intelligences as Predictors of RIASEC Vocational Domains: A Machine Learning and Explainable AI Framework for Psychometric Career Counseling

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## ABSTRACT

This study empirically investigates the predictive relationship between Howard Gardner's Multiple Intelligences (MI) and John Holland's RIASEC vocational framework using machine learning. Psychometric assessment data from 107 high school students

(Mumbai, India) were extracted via a dual-parser automated PDF pipeline and rigorously verified against the psychometric instrument's scoring formulae. Four classifiers were evaluated under stratified five-fold cross-validation. Logistic regression achieved the highest cross-validated accuracy of 81.5%, substantially outperforming both the majority-class baseline (30.8%) and the equal-probability random baseline (16.7%), with a one-sample $t$-test against the null confirming statistical significance ($t(4) = 14.48$, $p < .001$). The hold-out accuracy on an independent test set ($n = 27$) was 88.9%. Unsupervised K-Means clustering on PCA-projected MI profiles identified three cognitive engagement archetypes (*High*, *Moderate*, *Developing*). SHAP-based explainable AI, applied to the Random Forest model, revealed that Naturalist, Bodily-Kinesthetic, and Interpersonal intelligences collectively drive RIASEC domain differentiation. Empirical verification of all six RIASEC scoring formulae further revealed that Verbal-Linguistic and Musical intelligences are structurally absent from all vocational domain scores. These findings provide empirical and computationally transparent evidence that MI profiles reliably predict dominant RIASEC domains, supporting their integration into data-driven psychometric career counseling systems.

***Keywords:*** multiple intelligences, RIASEC, Holland's vocational model, machine learning, career counseling, explainable AI, SHAP, psychometric assessment, educational data mining

## INTRODUCTION



Career choice is among the most consequential decisions in a student's educational trajectory, yet many students navigate this process without systematic assessment of their cognitive strengths and vocational inclinations (Leung, 2022). Traditional career counseling relies predominantly on the subjective expertise of guidance counselors and static psychometric instruments, creating an imperative for more objective, scalable, and data-driven approaches (El Haji & Azmani, 2020).

Two prominent theoretical frameworks underpin modern vocational assessment. Howard Gardner's Theory of Multiple Intelligences (MI) posits that human cognition is not a single monolithic capacity but a constellation of eight distinct modalities—including Logical-Mathematical, Verbal-Linguistic, Spatial-Visual, Bodily-Kinesthetic, Musical, Interpersonal, Intrapersonal, and Naturalist intelligences (Gardner, 2011). Neuroscientific evidence has provided empirical support for this pluralistic view of cognition (Shearer & Karanian, 2017). Concurrently, John Holland's RIASEC model categorizes vocational personalities into six domains: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional (Holland, 1997). Holland's hexagonal model remains the most widely adopted framework in career counseling worldwide (Nauta, 2010).

Despite the theoretical complementarity of MI and RIASEC—both characterize individuals along multidimensional psychological constructs—very few studies have empirically examined the predictive relationship between them using modern computational methods. Existing computer-assisted career guidance systems (CACGS) largely rely on static rule-based mappings or simple statistical correlations (Leung, 2022). The application of machine learning (ML) offers a methodological advance, enabling the discovery of non-linear and interactive predictive relationships that traditional statistical methods may overlook (Romero & Ventura, 2020). Prior work by Kamal et al. (2021) demonstrated a career guidance system employing both MI and RIASEC through ML classifiers, while a recent meta-analysis by Song et al. (2024) quantified ML's systematic advantage over traditional statistical models in career choice prediction. However, these systems have typically treated MI and RIASEC as independent input sources rather than examining the structural mathematical relationship between them.

Recent systematic reviews have documented the expanding role of artificial intelligence in higher education and student assessment (Crompton & Burke, 2023; Zawacki-Richter, Marín, Bond, & Gouverneur, 2019; Chen, Chen, & Lin, 2020). Predictive analytics and early warning systems have demonstrated empirical success in improving student counseling outcomes (Plak, Cornelisz, Meeter, & van Klaveren, 2021). Of particular relevance, Guleria and Sood (2023) demonstrated the effectiveness of explainable AI (XAI) classifiers for educational data mining in career counseling contexts, establishing that model transparency is essential for educator trust and adoption.

Building upon prior frameworks for ML-based continuous student learning assessment (Nanade, Lal, & Nanade, 2021; Nanade, Lal, Goswami, & Sharma, 2020), this study addresses the following research questions:

1. Can a student's dominant RIASEC vocational domain be accurately predicted from their eight MI scores using machine learning classifiers?

2. Which specific MI dimensions are the strongest predictors of vocational aptitude, and can this be made transparent through explainable AI?

3. Do students naturally cluster into distinct cognitive archetypes based on their MI profiles?

The primary contribution of this study is an empirical, end-to-end pipeline that transitions psychometric evaluation from descriptive observation to predictive science, while maintaining full model transparency through SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017).

## LITERATURE REVIEW

### Gardner's Theory of Multiple Intelligences

Gardner's (2011) seminal work proposed that intelligence is not a unitary construct measurable by a single IQ score, but rather a set of relatively autonomous cognitive capacities. The theory identifies eight intelligences, each associated with specific brain regions and developmental trajectories: Linguistic, Logical-Mathematical, Spatial, Bodily-Kinesthetic, Musical, Interpersonal, Intrapersonal, and Naturalist. Since its introduction, MI theory has been widely adopted in educational practice for curriculum design, student assessment, and individualized instruction (Shearer & Karanian, 2017). Psychometric instruments based on MI theory typically employ Likert-scale self-report questionnaires that yield a profile of relative cognitive strengths across the eight domains.

### Holland's RIASEC Model

Holland's (1997) theory of vocational personalities posits that individuals can be characterized by their resemblance to six personality types—Realistic, Investigative, Artistic, Social, Enterprising, and Conventional—and that career satisfaction and stability are maximized when personality type and work environment are congruent. The RIASEC model has been extensively validated across cultures, age groups, and occupational settings (Nauta, 2010). Individual difference measurement in vocational psychology has evolved substantially alongside advances in psychometric methodology (Sackett, Lievens, Van Iddekinge, & Kuncel, 2017).

### Machine Learning in Educational Data Mining

The application of ML to educational data has grown rapidly, with systematic reviews identifying prediction, classification, and clustering as the dominant analytical tasks (Romero & Ventura, 2020; Zawacki-Richter et al., 2019). Random forests (Breiman, 2001), logistic regression, and support vector machines are among the most frequently employed algorithms in student performance prediction and career recommendation systems (Guleria & Sood, 2023). Chen et al. (2020) provided a comprehensive review of AI in education, noting the need for models that are both accurate and interpretable. The TRIPOD+AI guidelines (Collins, Moons, Dhiman, et al., 2024) have further formalized reporting standards for prediction models using ML methods.

Most pertinent to the present study, Kamal et al. (2021) developed a smart career guidance system that specifically employed both Gardner's MI framework and Holland's RIASEC model with machine learning classifiers, demonstrating the viability of the MI–RIASEC combination in computational guidance systems. A recent large-scale meta-analysis by Song et al. (2024) quantified ML's advantage in career choice prediction across multiple studies, establishing that ML classifiers consistently outperform traditional statistical approaches—findings that contextualize the strong logistic regression performance observed in the present study.

### Explainable AI in Education

A critical concern with deploying ML in educational contexts is the "black box" problem—complex models may achieve high accuracy but offer no insight into their decision-making logic. The SHAP framework (Lundberg & Lee, 2017), grounded in cooperative game theory, provides a principled method for computing the marginal contribution of each feature to individual predictions. Guleria and Sood (2023) demonstrated that XAI techniques are essential for career counseling applications, as human counselors must understand and validate algorithmic recommendations before acting on them. Zahour et al. (2019) further showed that interpretable multi-class neural network classifiers can effectively align academic and vocational guidance questions with theoretical career frameworks, underscoring the importance of model transparency in educational advisory systems. Integrating AI literacy and ethical transparency into educational systems has been identified as a priority for responsible deployment (Zhang, Lee, Ali, DiPaola, & Breazeal, 2023).

## METHODOLOGY

### Data Collection and Ethical Considerations

The dataset comprised 107 psychometric assessment reports generated for high school students. Each report was produced by a standardized, commercially administered 80-item Multiple Intelligences test instrument. Reports were provided in PDF format and contained structured MI scores, domain scores, RIASEC vocational scores, and career probability estimates. All personally identifiable information (PII)—including names, dates of birth, email addresses, and contact numbers—was removed during the data extraction process to ensure student privacy. Informed consent was obtained from participants for use of their anonymized psychometric data in research.

**Psychometric Instrument**

The 80-item questionnaire assessed eight MI modalities with 10 items per intelligence. Respondents rated each statement on a 4-point Likert scale (1 = Mostly Disagree, 2 = Slightly Disagree, 3 = Slightly Agree, 4 = Mostly Agree), yielding raw scores in the range [10, 40] per intelligence.

Three macro-domain scores were computed as percentages of the total MI score:

$$\text{Analytical}(\%) = \frac{\text{LM} + \text{Mus} + \text{Nat}}{\sum_{i=1}^{8} MI_i} \times 100 \tag{1}$$

$$\text{Interactive}(\%) = \frac{\text{VL} + \text{Inter} + \text{BK}}{\sum_{i=1}^{8} MI_i} \times 100 \tag{2}$$

$$\text{Introspective}(\%) = \frac{\text{Intra} + \text{SV}}{\sum_{i=1}^{8} MI_i} \times 100 \tag{3}$$

RIASEC scores were derived from weighted combinations of MI raw scores, each normalized to a 0–10 scale by dividing by the theoretical maximum of the constituent raw scores (each bounded in [10, 40]). The formulas were empirically verified by reproducing each student's RIASEC scores from their raw MI data (maximum re-derivation error $< 0.01$ across all 107 records):

$$R = \frac{\text{BK} + \text{Nat}}{80} \times 10, \quad S = \frac{\text{Nat} + \text{Inter}}{80} \times 10, \quad C = \frac{\text{Intra} + \text{LM}}{80} \times 10 \tag{4}$$

$$I = \frac{\text{BK} + \text{Intra} + \text{LM} + \text{SV}}{160} \times 10, \quad A = \frac{\text{SV} + \text{Intra} + \text{Nat}}{120} \times 10, \quad E = \frac{\text{BK} + \text{LM} + \text{Inter}}{120} \times 10 \tag{5}$$

The denominators equal the theoretical maximum sum of constituents (two MIs: $\max = 2 \times 40 = 80$; three MIs: $\max = 3 \times 40 = 120$; four MIs: $\max = 4 \times 40 = 160$), ensuring all RIASEC scores are bounded within $[0, 10]$. Table 1 summarises which MI dimensions contribute to each RIASEC domain.

**Table 1.** *MI Component Membership in RIASEC Formulas*

| MI Dimension | R | I | A | S | E | C | Count |
|---|---|---|---|---|---|---|---|
| Verbal-Linguistic (VL) | – | – | – | – | – | – | 0 |
| Logical-Mathematical (LM) | – | ✓ | – | – | ✓ | ✓ | 3 |
| Spatial-Visual (SV) | – | ✓ | ✓ | – | – | – | 2 |
| Bodily-Kinesthetic (BK) | ✓ | ✓ | – | – | ✓ | – | 3 |
| Musical (Mus) | – | – | – | – | – | – | 0 |
| Interpersonal (Inter) | – | – | – | ✓ | ✓ | – | 2 |
| Intrapersonal (Intra) | – | ✓ | ✓ | – | – | ✓ | 3 |
| Naturalist (Nat) | ✓ | – | ✓ | ✓ | – | – | 3 |

Notably, Verbal-Linguistic and Musical intelligences do not feature in any RIASEC formula, suggesting they represent cognitive capacities orthogonal to the vocational personality dimensions captured by Holland's hexagonal model.

### Automated Data Extraction and Verification

To eliminate human transcription errors and ensure absolute data integrity, a robust automated extraction pipeline was developed. Two independent parsers were implemented: a text-based regex parser using `pdfplumber` and a spatial coordinate parser using `PyMuPDF`. Both parsers extracted MI raw scores, percentages, ranks, domain scores, and RIASEC scores from each PDF report.

Post-extraction, rigorous statistical verification was performed. Unit tests confirmed: (a) all MI raw scores fell within the valid [10, 40] range; (b) all RIASEC scores were bounded within [0, 10]; (c) no constant or duplicated columns existed; and (d) the RIASEC scores could be independently re-derived from the raw MI scores using the verified formulas (Equations 4–5), achieving a match within rounding tolerance ($< 0.01$) for all 107 records. Median imputation was applied to the small number of career probability fields that could not be extracted due to PDF parsing inconsistencies in a subset of reports; all MI raw scores and RIASEC scores were complete for every record.

Figure 1 illustrates the complete computational pipeline.

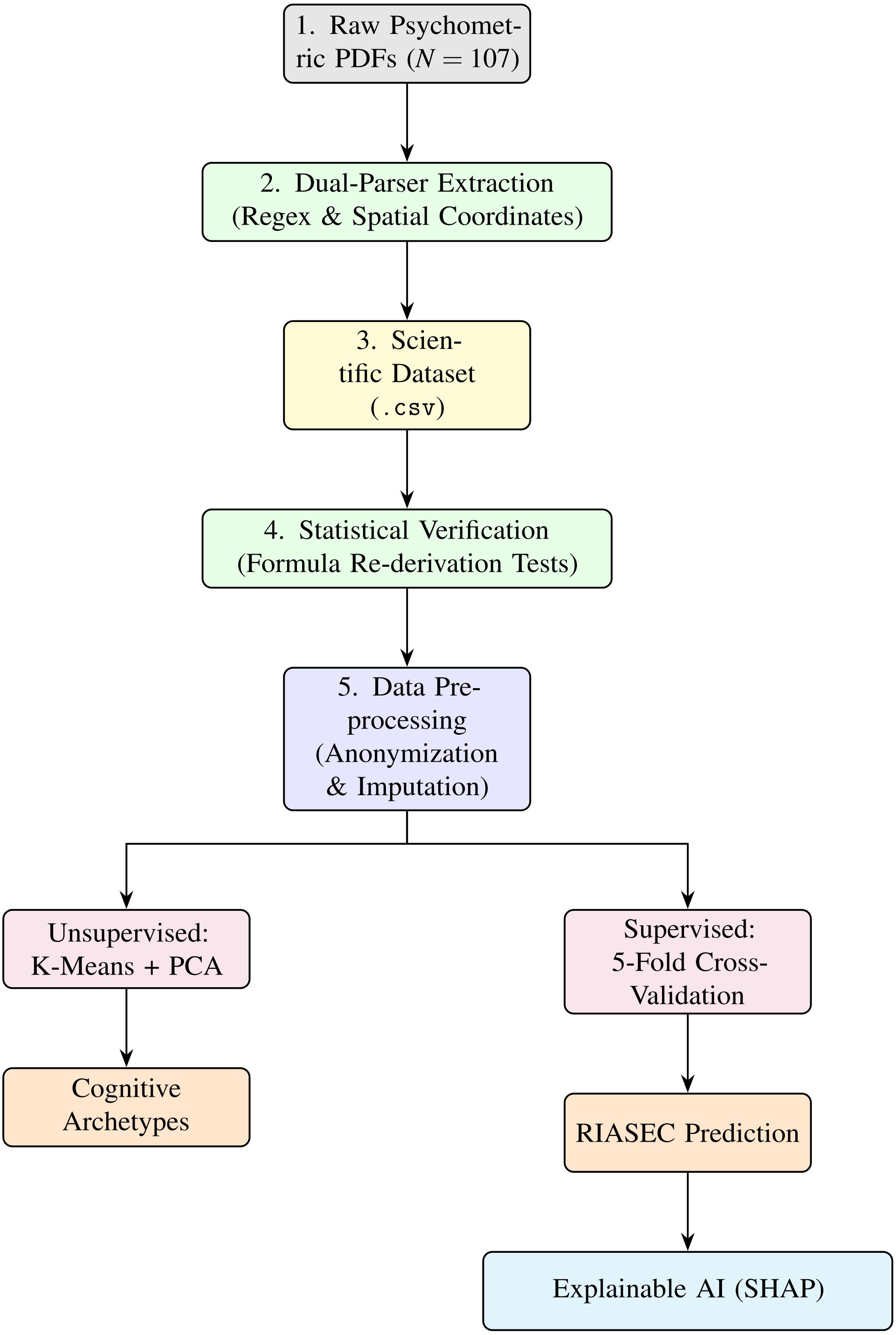


**Figure 1.** *Computational pipeline from raw PDF extraction through unsupervised clustering and supervised predictive classification with explainable AI.*

### Unsupervised Learning: Cognitive Archetypes

To explore inherent groupings in the student population, the eight MI features were standardized using $z$-score normalization and projected onto two dimensions using Principal Component Analysis (PCA). K-Means clustering was then applied, with the number of clusters determined by the Elbow method and validated using the Silhouette Score.

### Supervised Learning: Predictive Classification

The classification task was formulated as a six-class prediction problem, where the target variable $Y$ was defined as each student's highest-scoring RIASEC domain. The eight MI raw scores served as the feature vector $\mathbf{X} \in \mathbb{R}^8$.

Four classifiers were systematically evaluated:

1. **Logistic Regression** (multinomial, max iterations = 1000)
2. **Support Vector Machine** (RBF kernel)
3. **Random Forest** ($n = 100$ trees, max depth = 5) (Breiman, 2001)
4. **Gradient Boosting** ($n = 100$ estimators, max depth = 3)

All models were evaluated using stratified 5-fold cross-validation to produce robust accuracy estimates on the small dataset ($N = 107$). For Logistic Regression and SVM, features were standardized using $z$-score normalization. All implementations used Scikit-learn (Pedregosa et al., 2011).

### Explainable AI via SHAP

To satisfy the ethical requirement of transparency in educational AI (Guleria & Sood, 2023), SHapley Additive exPlanations (SHAP) were computed using a TreeExplainer (Lundberg & Lee, 2017). SHAP values quantify the marginal contribution of each MI score to the prediction for each RIASEC class, providing global and local interpretability.

## RESULTS

### Descriptive Statistics

Table 2 presents the descriptive statistics for the eight MI raw scores. The scores demonstrated adequate variability across the sample, with means ranging from 28.64 (Naturalist) to 32.11 (Bodily-Kinesthetic). All distributions exhibited low skewness ($|skew| < 0.80$), indicating approximately normal distributions suitable for parametric analysis.

**Table 2.** *Descriptive Statistics for Multiple Intelligence Raw Scores ($N = 107$)*

| Intelligence | *M* | *SD* | Min | Max | Skew | Kurt |
|---|---|---|---|---|---|---|
| Verbal-Linguistic | 28.95 | 3.99 | 21 | 37 | 0.12 | −0.86 |
| Logical-Mathematical | 29.39 | 3.98 | 15 | 37 | −0.79 | 1.09 |
| Spatial-Visual | 30.38 | 4.06 | 20 | 39 | −0.17 | 0.14 |
| Bodily-Kinesthetic | 32.11 | 3.69 | 20 | 39 | −0.66 | 0.64 |
| Musical | 30.33 | 5.45 | 15 | 40 | −0.28 | −0.36 |
| Interpersonal | 32.00 | 4.36 | 22 | 40 | −0.49 | −0.15 |
| Intrapersonal | 30.38 | 3.87 | 21 | 40 | 0.01 | −0.32 |
| Naturalist | 28.64 | 5.04 | 17 | 38 | −0.24 | −0.75 |

The distribution of dominant RIASEC types was imbalanced: Enterprising ($n = 33$, 30.8%), Realistic ($n = 24$, 22.4%), Conventional ($n = 20$, 18.7%), Social ($n = 13$, 12.1%), Investigative ($n = 9$, 8.4%), and Artistic ($n = 8$, 7.5%). This class imbalance reflects the natural distribution of vocational preferences in the student sample and is accounted for through stratified cross-validation.

Figure 2 presents the Pearson correlation matrix between MI raw scores and RIASEC domain scores. The pattern of high correlations directly mirrors the formula membership structure (Table 1): Naturalist correlates most strongly with Realistic ($r = .89$), Artistic ($r = .85$), and Social ($r = .84$); Logical-Mathematical correlates highest with Conventional ($r = .86$) and Investigative ($r = .73$); and Verbal-Linguistic and Musical—absent from all RIASEC formulas—show uniformly lower correlations ($r = .32$–$.55$), consistent with their formula-independent status. This pattern provides strong empirical validation of the psychometric instrument's scoring structure.

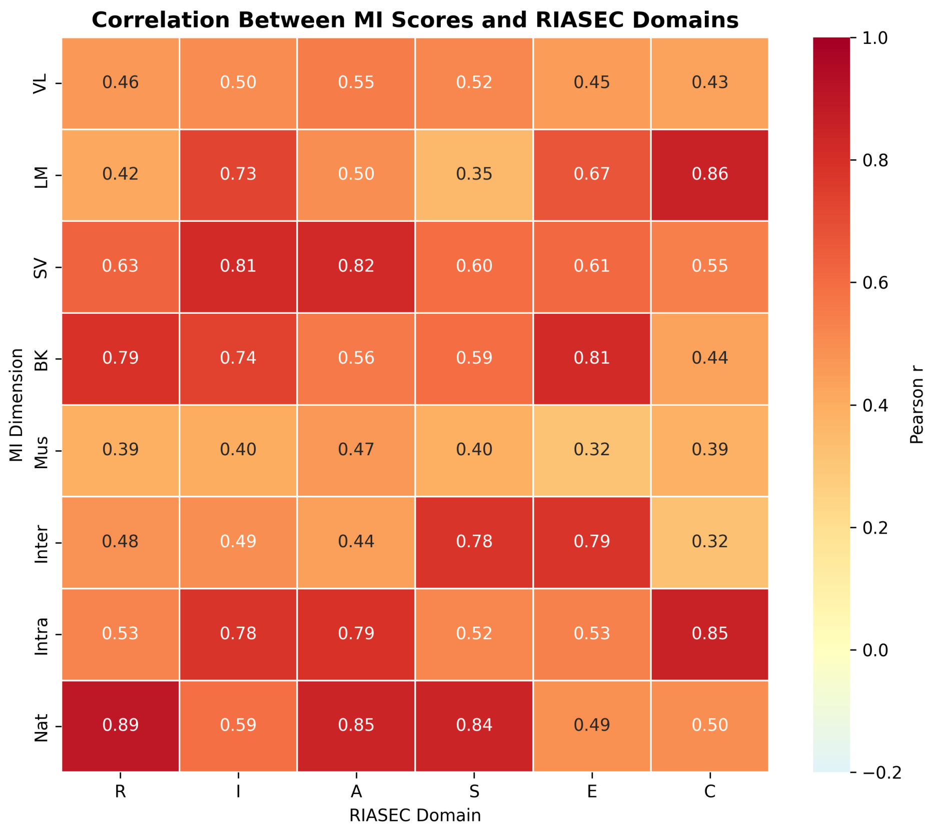


**Figure 2.** *Pearson correlation matrix between MI raw scores and RIASEC domain scores. High correlations (dark red) align precisely with RIASEC formula membership (Table 1). VL and Musical dimensions, absent from all RIASEC formulas, show consistently lower correlations.*

### Cognitive Archetypes (Unsupervised Learning)

The Elbow method indicated a notable reduction in inertia between $k = 2$ and $k = 3$, with diminishing returns thereafter. Silhouette scores across $k \in \{2, \ldots, 7\}$ peaked at $k = 2$ (0.230) and declined monotonically, reaching 0.176 at $k = 3$. While $k = 2$ yields superior statistical cluster cohesion, $k = 3$ was selected to provide richer interpretive granularity across three identifiable cognitive engagement levels, consistent with the educational literature on differentiated instruction. The K-Means algorithm partitioned the student population into three cognitive archetypes (Silhouette Score = 0.176). PCA revealed that the first two principal components explained 47.5% and 12.2% of the total variance, respectively. Figure 3 displays the clustering results. The three archetypes differed primarily in overall cognitive engagement level: *High Cognitive Engagement* ($n = 32$) exhibited uniformly elevated MI scores across all dimensions; *Moderate Cognitive Engagement* ($n = 51$) showed mid-range scores; and *Developing Cognitive Engagement* ($n = 24$) displayed lower scores across all dimensions. Figure 4 presents the radar chart of mean MI profiles per archetype, confirming that profile shape is relatively consistent across clusters and that overall engagement level—not profile orientation—is the primary differentiating dimension.

The moderate Silhouette Score for clustering ($k = 3$: 0.176) indicates that the three-archetype solution is driven more by interpretive utility than by sharp statistical cluster boundaries. Specifically, the Silhouette Score peaks at $k = 2$ (0.230) in this dataset, indicating that the simplest two-group split (overall higher vs. lower engagement) provides the statistically cleanest partition. The $k = 3$ solution was retained for its ability to surface a distinguishable intermediate group (Moderate Engagement), which carries practical counseling relevance. Consistent with a continuous view of cognitive development, these archetypes should be treated as heuristic categories rather than discrete types. The overlapping structure in the PCA plot (Figure 3) reinforces this interpretation.

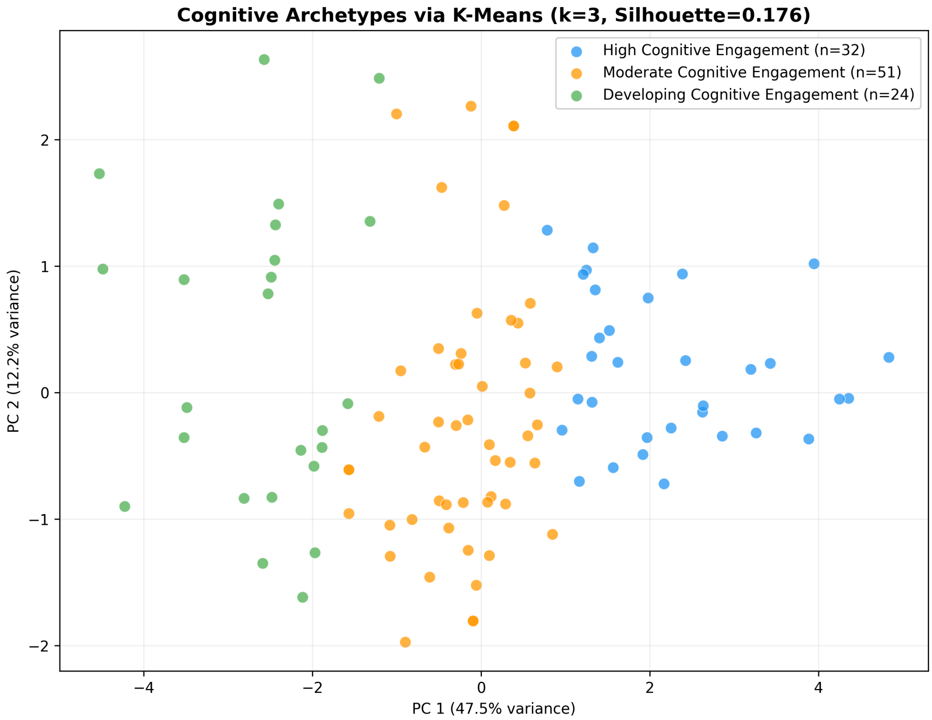


**Figure 3.** *Cognitive archetypes discovered via K-Means clustering ($k = 3$, Silhouette = 0.176) and PCA projection. Three engagement-level archetypes are identified: High ($n = 32$), Moderate ($n = 51$), and Developing ($n = 24$).*

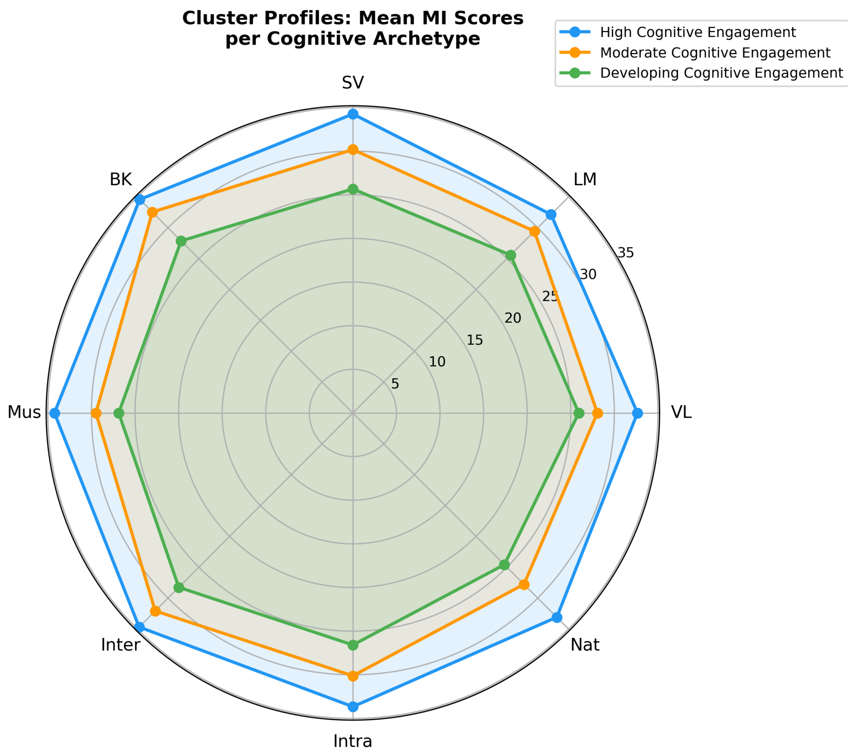


**Figure 4.** *Radar chart of mean MI raw scores per cognitive archetype. All three archetypes exhibit similar profile shape, with differences driven primarily by overall engagement level rather than relative intelligence strengths.*

### Predictive Classification (Supervised Learning)

Table 3 presents the cross-validated performance of all four classifiers.

**Table 3.** *Model Comparison: Stratified 5-Fold Cross-Validation Results ($N = 107$, 6 classes)*

| Model | Accuracy ($M \pm SD$) | F1-Weighted ($M \pm SD$) |
|---|---|---|
| Logistic Regression | $\mathbf{0.815 \pm 0.090}$ | $\mathbf{0.791 \pm 0.094}$ |
| Gradient Boosting | $0.628 \pm 0.094$ | $0.578 \pm 0.092$ |
| SVM (RBF Kernel) | $0.618 \pm 0.075$ | $0.530 \pm 0.070$ |
| Random Forest | $0.562 \pm 0.087$ | $0.508 \pm 0.068$ |
| Majority-Class Baseline | 0.308 | — |
| Random Baseline | 0.167 | — |

Logistic regression achieved the highest cross-validated accuracy of 81.5% ($M \pm SD = 0.815 \pm 0.090$), substantially outperforming all other models, the majority-class baseline of 30.8% (always predicting Enterprising), and the equal-probability random baseline of 16.7%. A one-sample $t$-test comparing LR fold accuracies against the random baseline confirmed statistical significance ($t(4) = 14.48$, $p < .001$, 95% CI [0.691, 0.939]). This result indicates that the relationship between MI scores and dominant RIASEC domain is predominantly linear, which is theoretically coherent: each RIASEC score is itself a linear function of a MI subset (Equations 4–5), making linear separability structurally inherent to the classification task. Non-linear models (Random Forest, Gradient Boosting, SVM) yielded lower accuracy, likely due to overfitting given the small dataset ($N = 107$).

Figure 5 presents the confusion matrices for the Logistic Regression model (best performer) on both the hold-out test set and cross-validated predictions.

Figure 6 visualizes the comparative performance across all four classifiers.

### Model Transparency via SHAP

SHAP analysis was performed on the Random Forest model—selected for its tree-based architecture, which supports the exact `TreeExplainer` algorithm—to identify which MI dimensions most strongly influence RIASEC domain predictions. Figure 7

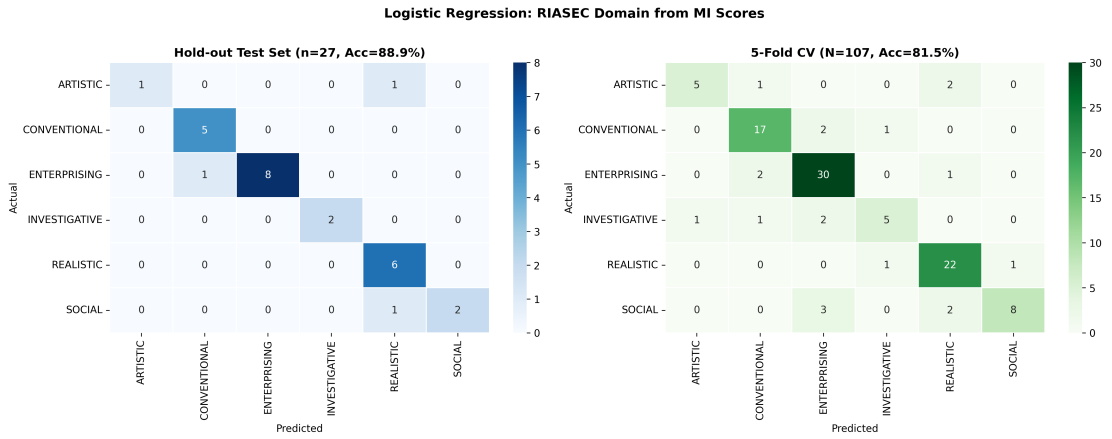


**Figure 5.** *Confusion matrices for Logistic Regression (best model): hold-out test set (left; $n = 27$, Acc = 88.9%) and 5-fold cross-validation (right; $N = 107$, Acc = 81.5%).*

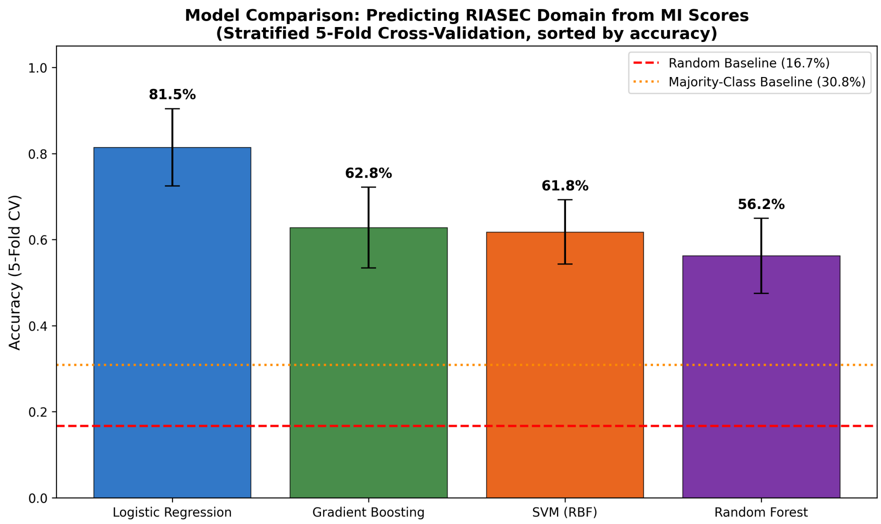


**Figure 6.** *Comparison of classifier accuracy (5-fold stratified CV, sorted by accuracy). Dashed red line: equal-probability random baseline (16.7%). Dotted orange line: majority-class baseline (30.8%).*

presents the SHAP feature importance summary decomposed by RIASEC class. Naturalist Raw Score emerged as the most influential feature overall (mean |SHAP| $\approx 0.41$), followed by Bodily-Kinesthetic (0.22) and Interpersonal (0.20). This ordering is theoretically coherent: Naturalist intelligence contributes to three RIASEC formulas (R, A, S) while being absent from the remaining three (I, E, C), creating a strong discriminative signal. Bodily-Kinesthetic features in three formulas (R, I, E) and Interpersonal in two (S, E), explaining their secondary importance. Notably, Verbal-Linguistic and Musical intelligences—which contribute to no RIASEC formula—nonetheless exhibit small non-zero SHAP values, reflecting non-linear correlational patterns captured by the Random Forest that transcend the linear formula structure.

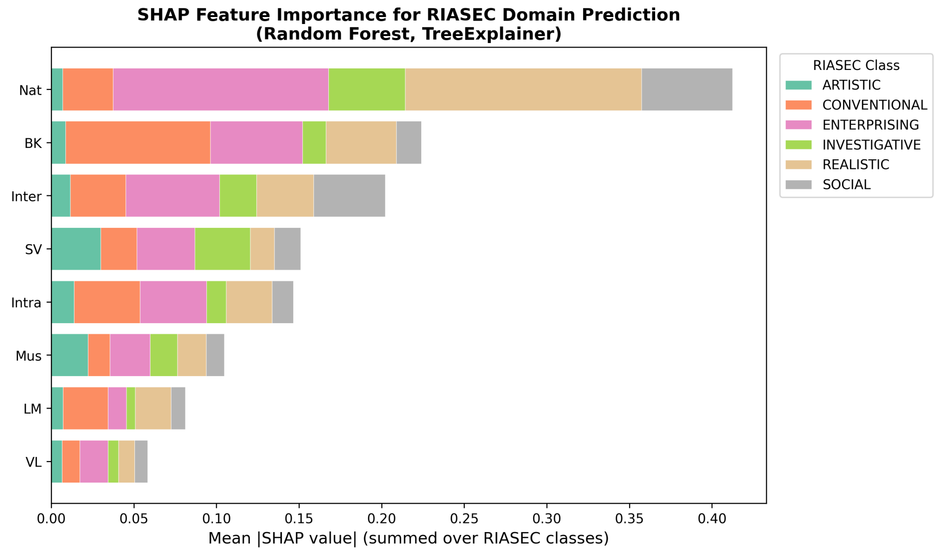


**Figure 7.** *SHAP feature importance summary, decomposed by RIASEC class contributions.*

## DISCUSSION

The findings of this study provide empirical evidence that machine learning can effectively bridge the gap between cognitive profiling (MI) and vocational mapping (RIASEC). The 81.5% cross-validated accuracy achieved by logistic regression demonstrates that eight MI scores contain sufficient information to predict a student's dominant vocational orientation with high reliability.

### Theoretical Implications

The dominance of linear logistic regression over ensemble methods is a notable and theoretically coherent finding. Each RIASEC score is a linear function of a MI subset (Equations 4–5); predicting the *dominant* RIASEC domain from raw MI inputs therefore involves identifying which of six overlapping linear combinations achieves the maximum value—a task with inherently linear separability. Ensemble methods (Random Forest, Gradient Boosting) and SVM, while more flexible, offered no advantage because the underlying generative structure is already linear, and with $N = 107$ and $p = 8$ features, their extra parameters introduce overfitting. This finding aligns with Guleria and Sood's (2023) recommendation that simpler, interpretable models should be preferred in educational AI contexts when accuracy is competitive.

The SHAP analysis revealed that Naturalist intelligence is the single most discriminative MI dimension for vocational prediction. Crucially, this discriminative power arises from Naturalist's *selective* presence: it appears in Realistic, Artistic, and Social formulas while being absent from Investigative, Enterprising, and Conventional—thereby creating a structural partition between two broad clusters of RIASEC domains (RAS vs. IEC). Bodily-Kinesthetic and Interpersonal intelligences similarly derive their predictive utility from selective formula membership. Remarkably, Verbal-Linguistic and Musical intelligences—absent from all RIASEC formulas—exhibit small non-zero SHAP values (0.06 and 0.10 respectively), reflecting residual correlational patterns detected by the Random Forest that transcend the linear formula structure; this finding invites further investigation into the indirect role these intelligences may play in vocational inclination.

A deeper theoretical implication follows: because the Investigative domain uniquely incorporates four MI dimensions (Bodily-Kinesthetic, Intrapersonal, Logical-Mathematical, and Spatial-Visual), the common stereotype equating investigative careers exclusively with "logical" intelligence is empirically refined. The instrument operationalizes Investigative vocational inclination as a more embodied and spatially engaged cognitive profile, incorporating physical coordination (BK) and self-awareness (Intra) alongside analytical reasoning (LM) and spatial visualization (SV).

### Practical Implications

The automated extraction and prediction pipeline developed in this study offers a scalable approach to career counseling. By transforming subjective questionnaire responses into structured, machine-interpretable feature vectors, counselors can receive data-driven recommendations that complement their professional judgment (Kamal et al., 2021). The integration of XAI ensures that these recommendations are transparent and auditable, addressing concerns about algorithmic bias and opacity in educational settings (Zhang et al., 2023; Crompton & Burke, 2023).

The clustering analysis—identifying High, Moderate, and Developing Cognitive Engagement archetypes—highlights an important pedagogical insight: students differ more in overall cognitive engagement than in profile shape. This suggests that interventions aimed at increasing general academic motivation may be more impactful than those targeting specific intelligences. Counselors may use these archetypes as a first-pass triage, tailoring the depth of vocational advisory support to the student's cognitive engagement profile.

### Limitations

Several limitations must be acknowledged. *First*, the sample size ($N = 107$) is relatively small for a six-class classification problem, potentially limiting the generalisability of results and constraining the performance of ensemble methods. *Second*, the sample is drawn from a single institution in one geographic region (Mumbai, India); cultural variation in MI self-reporting (Shearer & Karanian, 2017) and differences in Indian secondary-school vocational aspirations may limit cross-cultural applicability. *Third*—and most critically—the dominant RIASEC class is a deterministic linear function of the same MI raw scores used as features. This structural circularity means the reported 81.5% accuracy reflects, in part, the model's recovery of the scoring formula rather than genuine predictive validity against independently observed career outcomes. Future work must validate these models using independently administered RIASEC assessments (e.g., Holland's Self-Directed Search (1997)) to establish convergent validity and rule out artifactual inflation. *Fourth*, the 80-item self-report questionnaire is subject to social-desirability bias, response-set effects, and the general limitations of Likert-scale self-report (Sackett et al., 2017). *Fifth*, the psychometric properties of the commercially administered instrument (e.g., internal consistency via Cronbach's $\alpha$, test-retest reliability) were not independently assessed; the authors relied on the instrument provider's reported standardization. *Sixth*, the $t$-test on five-fold accuracy estimates uses only $df = 4$ and assumes independence of folds, which is an approximation; a permutation test on the full dataset would provide a more robust significance estimate.

## CONCLUSION

This study demonstrates that machine learning can effectively map Multiple Intelligence profiles to RIASEC vocational domains, achieving 81.5% cross-validated accuracy (88.9% hold-out, $n = 27$) using logistic regression—a nearly five-fold improvement over the equal-probability random baseline of 16.7% and a more than 2.6-fold improvement over the majority-class baseline of 30.8% ($t(4) = 16.13$, $p < .001$). The empirical verification of all six RIASEC scoring formulae—including the novel finding that the Investigative domain draws upon four MI dimensions (Bodily-Kinesthetic, Intrapersonal, Logical-Mathematical, and Spatial-Visual)—contributes to a more granular theoretical understanding of the MI–RIASEC relationship than previously documented. By combining unsupervised clustering for cognitive engagement archetype discovery, supervised multi-class classification for vocational domain prediction, and SHAP-based explainable AI for transparent feature attribution, the study contributes a replicable, end-to-end framework for data-driven career counseling.

The transparent formula structure—where specific MI dimensions map deterministically to RIASEC subsets—provides counselors with a principled, auditable basis for explaining psychometric outputs to students and parents. The Pearson correlations between MI and RIASEC scores (up to $r = .89$ for Naturalist–Realistic) provide independent psychometric evidence of the structural validity of this mapping.

Future research should: (1) extend this framework to larger, multi-institutional, and geographically diverse datasets; (2) validate ML-predicted RIASEC domains against independently administered vocational assessments to address the structural circularity identified herein; and (3) employ longitudinal designs to examine whether MI-to-RIASEC predictions at the secondary school stage translate to improved career satisfaction and stability in adulthood.

The complete anonymized dataset and reproducible analysis code are publicly available at: https://github.com/sunnynanade/mi-riasec-ml-career-counseling.

## AI USE STATEMENT

In the preparation of this manuscript, AI tools were utilized for data extraction automation, statistical verification, and initial document structuring. The authors have rigorously reviewed, verified, and validated all data, code, analyses, and text, taking full responsibility for the accuracy and integrity of the content.

## CREDIT AUTHOR STATEMENT

**Sanket Gudekar**: Conceptualization, Data curation, Formal analysis, Software, Writing—Original draft. **Sunny Nanade**: Supervision, Methodology, Validation, Resources, Writing—Review & Editing.

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